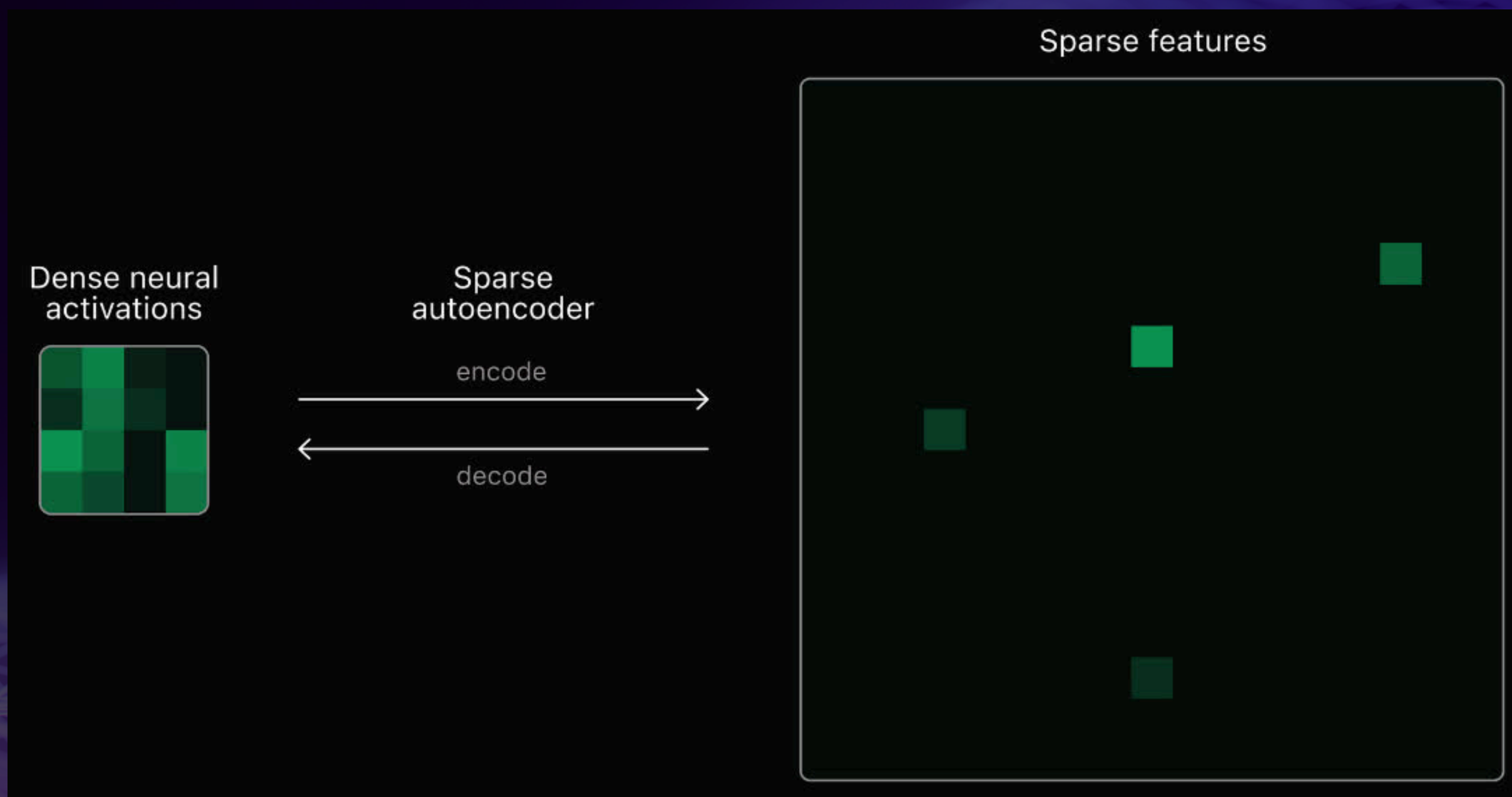


Tokenization and Paraphrase Sensitivity in Gemma 3 Using SAE Features



Experimental Setup

Dataset: azhx/counterfact

Model: google/gemma-3-4b-pt

SAEs: google/gemma-scope-2-4b-pt

Extracted representation:

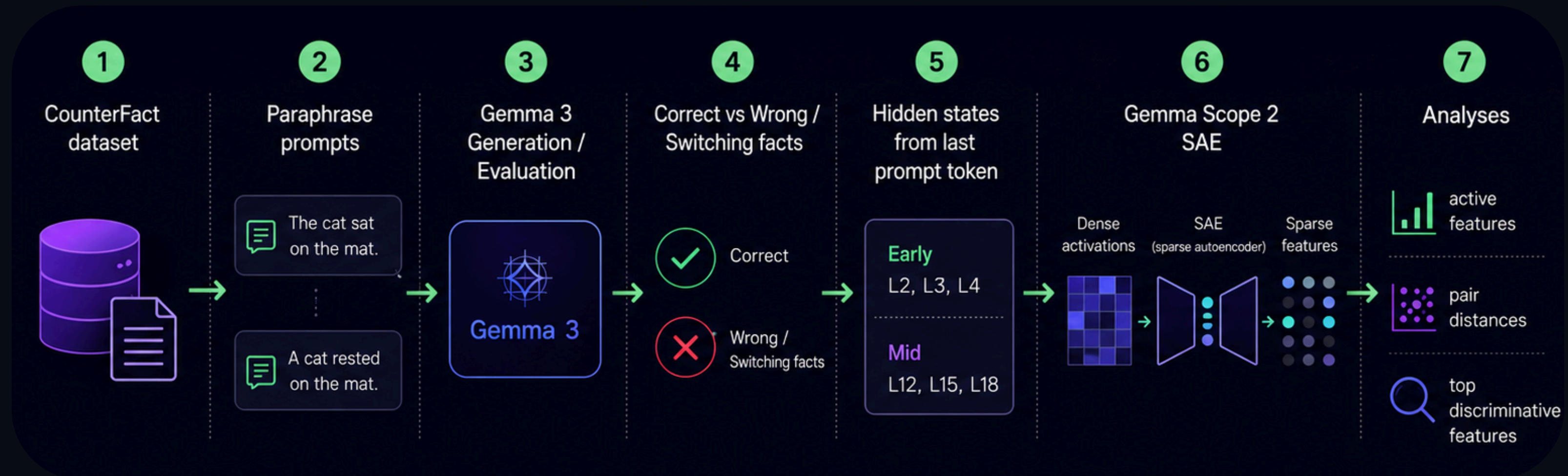
last prompt token hidden state

Input: two paraphrase prompts per fact

Output: label: correct / wrong answer

Layers:

- 2, 3, 4
- 12, 15, 18



Pipeline

1. The pipeline starts with CounterFact paraphrases,
2. runs Gemma 3 to identify correctness switches,
3. extracts hidden states,
4. converts them into sparse SAE activations,
5. Then tests if switching facts show larger internal representation shifts than stable facts



SCRIPTS

1) create_counterfact_prompt_variants.py

Input Dataset Preaperation

- Loads CounterFact and its paraphrase prompts.
- Created fact IDs from case IDs and assign to each fact and paraphrase.
- Saves a CSV with one row per prompt.
- This CSV is the input for Gemma-3 generation.

The dataset has:

- 2191 facts
- 4382 prompt rows
- 2 paraphrases per fact

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Gemma result:

- Correct rows: 1343
- Wrong rows: 3039
- Accuracy: about 30.65%

Prompt Dataset for Gemma

Output of the current script

fact_id	subject	paraphrase_variant	prompt_given_to_model	true_answer
cf_20952	Angola	paraphrase_00	Plaque - located at Punkeydoodle's Corners. Angola belongs to the continent of	Africa
cf_20952	Angola	paraphrase_01	Last updated on 4 February 2017. Angola is a part of the continent of	Africa
cf_3648	Shanghai	paraphrase_00	Hamid Berhili (MAR), 10:4 Paul Ingle (GBR) def. The twin city of Shanghai is	Barcelona
cf_3648	Shanghai	paraphrase_01	Palgrave Macmillan, 2013. Shanghai is a twin city of	Barcelona

SCRIPTS

2) run_counterfact_gemma3.py

Feeds each paraphrase prompt into Gemma 3, saves the generated continuation, and applies strict correctness grading.

Strict grading logic:

- Uses the real factual answer: correct_answer / target_true
- Normalizes capitalization and punctuation
- Example: Africa. → africa
- Checks only the beginning of Gemma’s continuation(first 12 words)
- Marks the row correct if the true answer appears

Gemma input prompts and generation outputs preview

fact_id	subject	paraphrase_variant	prompt_given_to_gemma	true_answer_used_for_grading	gemma_generated_continuation	strict_matched_answer	is_correct
cf_20952	Angola	paraphrase_00	Plaque - located at Punkeydoodle's Corners. Angola belongs to the continent of	Africa	North America. It is located in the state of New York. The population of	N/A	False
cf_20952	Angola	paraphrase_01	Last updated on 4 February 2017. Angola is a part of the continent of	Africa	Africa. It is a country that is located in the south of the continent.	Africa	True

SCRIPTS

3) create_counterfact_switching_subset.py

Groups the two paraphrase outputs belonging to the same fact.

```
correct_correct = both paraphrases correct
wrong_wrong     = both paraphrases wrong
correct_wrong   = one correct, one wrong
```

The correct_wrong group is the main **switching subset**

RESULT

Total rows: 4382
Total facts: 2191
Correct rows: 1343
Wrong rows: 3039

=== Pair Type Breakdown ===
correct_correct : 401 facts (18.3%)
correct_wrong : 541 facts (24.7%)
wrong_wrong : 1249 facts (57.0%)

Switching facts: 541

Gemma input prompts and generation outputs preview

fact_id	subject	paraphrase_variant	prompt_given_to_gemma	true_answer_used_for_grading	gemma_generated_continuation	strict_matched_answer	is_correct
cf_20952	Angola	paraphrase_00	Plaque - located at Punkeydoodle's Corners. Angola belongs to the continent of	Africa	North America. It is located in the state of New York. The population of	N/A	False
cf_20952	Angola	paraphrase_01	Last updated on 4 February 2017. Angola is a part of the continent of	Africa	Africa. It is a country that is located in the south of the continent.	Africa	True

SCRIPTS

4) extract_hidden_states_gemma3.py

Run every prompt through Gemma 3 again but this time instead of generating text, extract the model’s hidden states(vectors).

- Layers selected: 2,3,4 and 12,15,18
- Hidden state at the last prompt token is used (the internal representation right before the model predicts the answer)

hidden_state_layer_norm_summary.

layer	hf_hidden_state_index	pair_type	n_prompts	n_facts	correct_rate	mean_n_tokens	mean_l2_norm	std_l2_norm
2	3	correct_correct	802	401	1.0	20.65	1192.4066	99.5988
2	3	correct_wrong	1082	541	0.5	20.24	1195.7648	101.6038
2	3	wrong_wrong	2498	1249	0.0	20.01	1203.4535	105.1995
3	4	correct_correct	802	401	1.0	20.65	1281.1035	125.4658
3	4	correct_wrong	1082	541	0.5	20.24	1293.786	150.2971
3	4	wrong_wrong	2498	1249	0.0	20.01	1299.718	152.7556
4	5	correct_correct	802	401	1.0	20.65	2708.409	246.5733

- layer: Gemma layer used for extraction
- hf_hidden_state_index: matching Hugging Face hidden-state index
- pair_type: correct-correct, correct-wrong, or wrong-wrong group
- n_prompts: number of prompt rows
- n_facts: number of facts
- correct_rate: group correctness rate
- 1.0 = both paraphrases correct
- 0.5 = switching fact
- 0.0 = both paraphrases wrong
- mean_n_tokens: average prompt length
- mean_l2_norm: average hidden-state vector size
- std_l2_norm: variation in vector size

RESULT

Input rows: 4382
Layers: 6
Hidden dimension: 2560 (Gemma 3 4B)
Activation tensor shape: (4382, 6, 2560)

4382 prompts × 6 selected layers × 2560 hidden dimensions

SCRIPTS

5) extract_sae_features_gemma3_manual.py

After extracting dense hidden states from Gemma, pass those states through Gemma Scope sparse autoencoders. The goal is to convert each 2560-dimensional dense vector into a sparse feature activation vector.

Process

- Loads the matching Gemma Scope SAE for each selected layer.
- Uses layers: 2, 3, 4, 12, 15, 18
- Each SAE has width 16384 features.
- Applies the JumpReLU encoder to convert dense hidden states into sparse feature activations.



sae_top_features_selected_layers.md

layer	feature_id	active_count	active_fraction	total_activation	mean_activation_when_active	max_activation
2	9020	2540	0.5796	81239.9066	31.9842	69.6964
2	749	1170	0.267	496296.113	424.1847	480.8817
2	895	828	0.189	61681.5698	74.4946	156.5026

The most frequently active SAE features for selected layers across all 4382 prompts.

- Each feature_id is one SAE feature.
- active_count shows how often the feature activated across prompts.
- active_fraction shows the fraction of prompts where it appeared.
- Activation columns show how strongly the feature activated.
- total_activation: Sum of activation strength over all prompts

RESULT

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Mean active features by layer:

layer	
2	6.175034
3	8.230488
4	9.988818
12	17.058421
15	16.962346
18	16.755363

SCRIPTS

6) [scripts/analyze_sae_features_counterfact.py](#)

analyzes the SAE activations by correctness group. It compares correct vs wrong prompts and switching vs non-switching facts, then identifies SAE features that are more associated with correct or wrong generations

SAE activations difference between the correct and the wrong paraphrase(switching facts)

layer	is_correct	n_rows	mean_active_features	std_active_features	mean_sum_activation	mean_max_activation
2	False	541	7.231053604436229	5.4900783067081935	929.6996492114394	373.6169810409687
2	True	541	6.044362292051756	4.656921432820652	825.0866702039228	375.095483466129
3	False	541	9.306839186691313	5.656373772127962	1121.7044623581187	402.0554327268477
3	True	541	7.927911275415896	4.5068237439221	994.6152479132971	412.9589193347642
4	False	541	10.578558225508319	3.977695050834949	1375.3751432803112	448.7640467730115
4	True	541	9.537892791127542	3.356716097446332	1271.007871842869	451.62564549472546
12	False	541	17.03327171903882	4.77675721187682	4374.847916185305	958.035771001509
12	True	541	17.42513863216266	4.226069471174126	4476.540273275041	990.5423084195572
15	False	541	17.382624768946396	4.7116476383745205	5200.374286080464	874.7196823373961
15	True	541	16.961182994454713	4.887282218863707	4921.574740425716	847.8004201159239
18	False	541	16.99260628465804	4.82585061109719	8601.686883556204	1741.6532956051078
18	True	541	16.720887245841034	4.356105543674685	8538.147954634069	1719.9965033963074

RESULT

Correct and wrong switching prompts differ slightly in SAE activation patterns, but not in a simple global way. The more important signal is likely in which features activate, not just how many.

RESULT

This motivates the next analysis: instead of only asking how many features activate, I will compare which features activate and how far apart the two paraphrases are in SAE feature space.

SCRIPTS

7) analyze_sae_pair_distances.py

compare the two paraphrases of the same fact, at the same layer, in SAE feature space(sae feature activations vector). To checks whether switching facts have larger internal representation shifts than stable facts.
For each fact and layer, it compares: paraphrase_00 vs paraphrase_01
using distances like:

- cosine distance (to compare the direction of the full SAE activation vectors)
- L2 distance (to measure absolute difference in activation strength)
- Jaccard distance (to see if two paraphrases activate the same SAE feature)

sae_pair_distance_summary

Example layer 18:

pair_type	mean_cosine	mean_jaccard	mean_shared
correct_correct	0.399	0.696	6.91
correct_wrong	0.603	0.805	4.85
wrong_wrong	0.588	0.789	5.25

$$D_{\text{correct_wrong}} > D_{\text{wrong_wrong}} > D_{\text{correct_correct}}$$

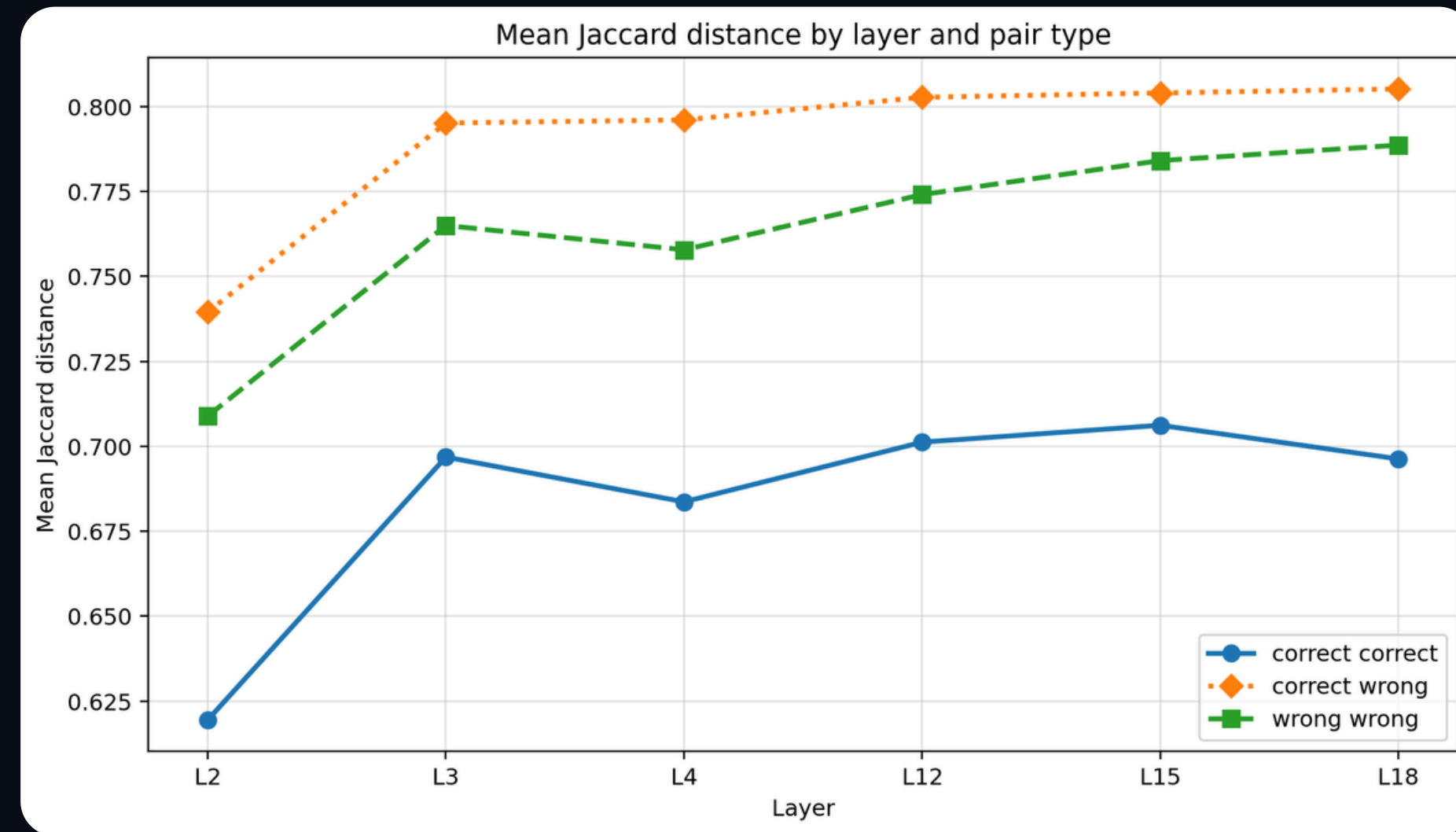
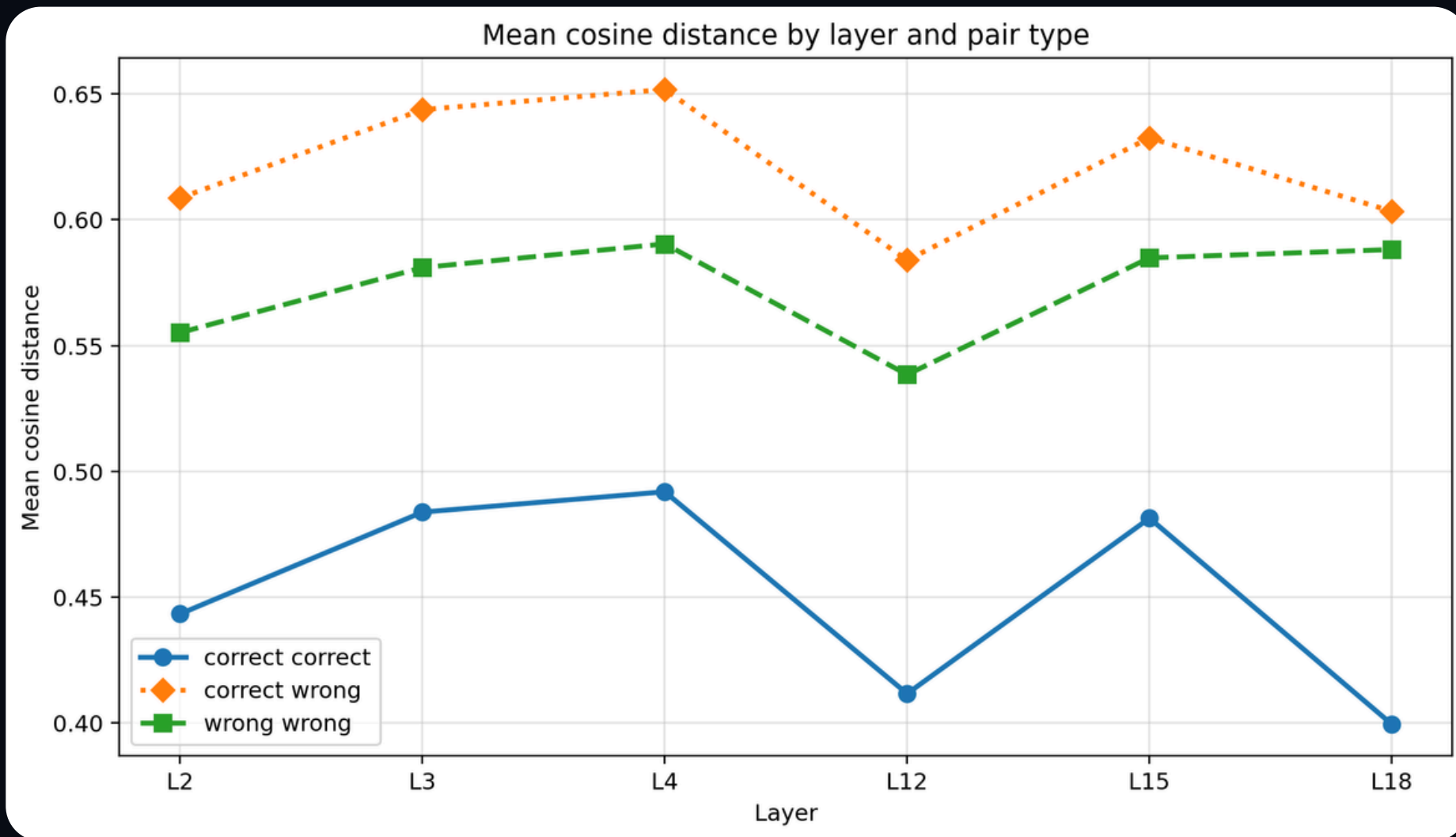
RESULT

For almost every layer, the correct_wrong pairs have the largest distances between the two paraphrases.

Switching facts show larger SAE feature-space distances between paraphrases than stable-correct facts.

This suggests that when a paraphrase changes Gemma's correctness, it is also associated with a larger internal representation shift.

Correct-Wrong Pair Graphs



Conclusion (So far)

Paraphrases that change Gemma's factual correctness are associated with larger shifts in sparse SAE feature space.